

AADARSH K A S

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Career Objective

I'm a final-year hardware engineering student who approaches engineering by starting with the problem, understanding the system around it, and building toward a practical solution. I work across RTL, digital architecture, embedded systems, and real-world products, while thinking about technology, execution, users, and purpose together. I learn by building, refine through execution, and aim to turn ideas into practical systems that solve meaningful problems.

Technical Skills & Tools

- **HDL / Languages:** Verilog (RTL Design, Simulation, Synthesis), C, Embedded C, Python
- **EDA Industry:** Synopsys VCS, Verdi, Fusion Compiler, Design Compiler, Cadence Virtuoso
- **EDA Open Source:** OpenROAD, OpenLane 2, Yosys, Icarus Verilog, GTKWave, WaveDrom
- **FPGA Tools:** AMD Vivado, Intel Quartus Prime
- **PCB / Simulation:** Altium Designer, EasyEDA, LTSpice, Multisim
- **Platforms:** Altera DE2, Renesas Forge SLG47910 (Vicharak Shira Lite), Raspberry Pi, ESP32, RP2040, Arduino
- **OS / Tools:** Linux, Git, GitHub

Internship

- Apr – May 2026 **Student Intern – Digital Design,** [Struent Semiconductors](#)
Chennai, India
 - Studying and applying digital circuit design fundamentals combinational/sequential logic, hardware integration, RTL coding under industry mentorship using professional EDA tools.
- Jun – Sep 2025 **Drone Engineer Intern,** [Big Bang Boom Solutions](#)
Chennai, India
 - Electronics integration, assembly, sensor calibration, and field testing for defence-grade UAV platforms, validating flight stability and mission readiness.
- Jan – Jun 2025 **R&D Intern,** [Spinacle Technologies Pvt. Ltd.](#)
Chennai, India
 - Designed power/area-optimized multi-layer PCBs in Altium Designer and developed ESP32 firmware in Embedded C for a healthcare IoT MVP enabling real-time multi-sensor data acquisition.
- Apr – May 2024 **AIoT Intern,** [Spinacle Technologies Pvt. Ltd.](#)
Chennai, India
 - Built an ESP32-based audio monitoring system integrated with Microsoft Azure, reducing alert latency by 10% vs. prior pipeline.

Projects

- SkyWater 130nm
OpenLane 2 **Custom 130nm CMOS NOT Gate, RTL-to-GDSII**
 - DRC/LVS-clean GDSII layout via OpenLane 2 (Yosys + OpenROAD); 50×50 μm footprint with West-In/East-Out pin ordering for modular hierarchical integration.
- Renesas Forge
SLG47910 FPGA **On-Chip Tsetlin Machine, Edge-ML HW/SW Co-Design** [To Learn More\[↗\]](#)
 - Developed a Verilog-2001 FPGA binary classifier using LFSR-gated stochastic logic and 12-clause voting, with an RP2040 MicroPython supervisor for GPIO-based training and autonomous convergence.
- Altera DE2
Verilog RTL **Hardware-Driven Pong, FPGA Game Engine** [To Learn More\[↗\]](#)
 - Pure RTL Pong no CPU, no software with 640×480 VGA output, real-time physics, collision detection, score tracking, and synchronous game state machine.
- Wearable
Multi-Sensor Fusion **Mento, Predicts injuries & Prevent It.** [To Learn More\[↗\]](#)
 - Mento is a real-time wearable system that predicts sports injuries before they happen using multi-sensor fusion, edge computing, and an LSTM trained on continuous biometric streams.
- Embedded C
EMG / DSP **MuscleSync, Real-Time EMG-Based Control System.** [To Learn More\[↗\]](#)
 - Embedded C firmware for EMG signal acquisition and digital filtering; 85% pattern-recognition accuracy for assistive control applications.
- Raspberry Pi
Cloud AI **Gut Health Management System, Non-Invasive Diagnostics.** [To Learn More\[↗\]](#)
 - Built a non-invasive gut health monitoring system using Raspberry Pi; captured multimodal data (image + audio), processed via cloud AI, achieving 80% diagnostic detection accuracy.

Educational Qualification

2023 – 2027 Final Year	B.E. – Electronics Engineering (VLSI Design & Technology) <i>Rajalakshmi Institute of Technology, Chennai</i> Coursework: VLSI Design, Digital Systems, Verilog HDL, Microprocessors & Microcontrollers, Embedded Systems, Design Thinking	CGPA: 7.93 / 10
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Certifications

• RTL, IP Integration & SoC Signoff — NIELIT • Introduction to RISC-V — The Linux Foundation • From Basics to GenAI Practice — NVIDIA • AWS IoT — Amazon Web Services	• PCB Design — Altium Education • Digital Circuits — NPTEL • Data Visualization — Kaggle
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Achievements

• Second Runner-Up, represented Team Atomic Minds at the ANNA Incubator × IIT PALS Pitch Fest 2026, recognized in a Dinamalar news article LINK TO ARTICLE • Runner-up, MedTech Track IIT PALS InnoWAH 2026, for a clinically focused hardware solution. • TOP 5 National Finalist, Smart India Hackathon 2026 designed a wearable targeting the sports & health sector. • Round-1 Qualifier, Lam Research Challenge a nationwide systems-engineering competition hosted by IISc Bangalore, T-Works, and Lam Research. • Top 100 Team nationwide in the 1-TOPS Program A Government initiative for One Tape Out Per Student. • Published IEEE author co-authored “Smart Wheelchair Based on Voice Recognition for Physically Disabled People” (ICPECTS 2024).
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Co-Curricular Activities

March 2026 - Present Rit	Student Member , Center of Excellence for Semiconductor Design • Learning and understanding the Semiconductor Design and the practicing the industry EDA tools sponsored by C2S.
Aug 2025 – Feb 2026 RIT	Student Co-Ordinator , <i>Center for Advanced Data Analytics</i> • Lead student projects and mentor peer learners in data collection and project development.
Feb – Aug 2025 RIT	Council Member , <i>Center of Excellence for IoT</i> • Guided 50+ students through hands-on IoT project development from prototyping to implementation.
Sep 2024 – Oct 2025 RIT	IoT Expert , <i>Google Developer Groups on Campus</i> • Led IoT projects and knowledge-sharing workshops for the campus developer community.